Experiments Draft: Embodied Reasoning as Grounded Selection

Anonymous Submission
Anonymous Affiliation

July 26, 2026

Experiments

Embodied Reasoning

Evaluation protocol and evidential order. We evaluate at anticipation points from the Ego4D Goal-Step held-out split. The frozen world model receives only the observation and proposes a shuffled Top-K action boundary; its scores and ranks are hidden from the policy. The policy receives that boundary together with the observation and completed-action history. Unless noted otherwise, all comparisons use the same first-person cohort and common covered set ($a_t^{\mathrm{GT}} \in D_t$; n=1,000). We order the evidence from outcome to mechanism: controlled selection, capability decomposition, and paired interventions on history and task belief. This separates a change in *what is selected* from a superficial change in *how the trace is written.*

Candidate Alignment, Not Answer-Only Supervision, Improves Selection

The primary comparison controls the backbone, examples, optimization budget, and evaluation prompt; only access to the candidate set during training changes. Candidate alignment outperforms answer-only supervision by $+7.70$ pp on the official paired set (95% CI $[+3.73, +11.93]$). The answer-only control remains indistinguishable from the unaligned policy, whereas candidate alignment also clears the frozen world-model Top-1 reference (paired CI lower bound $+0.87$ pp). Because label exposure and training budget are matched, this contrast isolates learning to discriminate among grounded alternatives, not simply seeing more correct answers. Hidden world-model scores further rule out direct rank copying. Figure 1 summarizes where the resulting capability gains appear.

Where the Gain Appears: A Capability Profile

Aggregate accuracy does not reveal whether a policy preserves a correct world-model prior or repairs an incorrect one. We therefore separate G_1 retention from G_2 correction (Figure 1; Table 1). Candidate alignment improves both, whereas answer-only supervision weakens retention while offering little correction. The final EGO model shifts the profile further toward retaining valid priors without uniformly improving correction. Thus, providing a candidate boundary at inference is insufficient: training must teach the policy when to trust its leading hypothesis and when trajectory context warrants an override. The decomposition also prevents the aggregate gain from being attributed to a single easy subset.

Trajectory continuity exhibits the same structure. Candidate alignment recovers substantially more true continuations, and the final EGO model achieves the strongest continuation recall. Precision remains broadly stable across conditions, so the recall gain is not explained by a looser tendency to repeat the previous action. Instead, training appears to improve access to an already reliable temporal cue.

Causal Test: The Learned Advantage Is Carried by History

We next remove completed-action history while holding the observation, candidate set, and model fixed. Candidate alignment loses 10.1 pp (95% CI $[+6.8, +13.3]$), and its advantage over both controls disappears. The replay-stage model shows the same qualitative dependence, although this intervention was not repeated after grounding oversampling. A better static action prior would survive removal of history; the observed collapse instead identifies trajectory use as the source of the candidate-alignment gain. This is the strongest evidence that the policy interprets the world-model boundary conditionally rather than merely adopting a different answer frequency.

Useful Evidence, Not More Evidence

We finally ask whether a generated trace contains decision-relevant state or only persuasive wording. Answer-only supervision makes prior-action language more frequent but less predictive of correctness. Candidate alignment preserves a strong association between cited temporal evidence and successful selection, while the final EGO model uses such language more selectively and with the highest conditional utility (Figure 1). The relevant distinction is therefore not whether a model *says* that it used history, but whether

	Grounded selection			Trajectory continuity		Evidence-conditioned selection	
	SelAcc @Top-K	G_1: retain WM-correct	G_2: correct WM-wrong	Continuation recall	Continuation precision	Evidence mentioned	Evidence absent
Base	21.0	29.7	18.7	28.4	72.7	31.8	18.4
GT-only	21.9 +0.9	26.5 -3.2	20.8 +2.1	32.9 +4.5	71.3 -1.4	24.1 -7.7	21.0 +2.6
Cand.-CE	28.8 +7.8	37.4 +7.7	27.8 +9.1	43.5 +15.1	71.1 -1.6	38.4 +6.6	26.1 +7.7
EGO	28.8 +7.8	42.5 +12.8	25.6 +6.9	48.4 +20.0	68.2 -4.5	44.7 +12.9	26.4 +8.0

Δ vs. Base (pp) -20 0 $+20$

Figure 1: **Capability profile across controlled training conditions.** Each cell reports the raw percentage on the common covered set (n=1,000), followed by its signed change from the Base row. Color encodes that change on a shared ±20-percentage-point diverging scale (blue above Base, orange below), so saturation is proportional to effect size. Candidate alignment gains broadly in grounded selection and continuation recall, whereas answer-only supervision (GT-only) does not. EGO strengthens retention and continuation recovery further, but not on every axis. The evidence-conditioned columns are descriptive associations, not interventions.

Metric	What it tests
SelAcc@Top-K	Whether the policy selects the ground-truth action from the shuffled world-model Top-K candidate set when scores and ranks are hidden.
G_1: retain WM-correct	Whether the policy preserves a valid predictive prior when the world-model Top-1 action is correct.
G_2: correct WM-wrong	Whether the policy overrides an incorrect world-model Top-1 prediction and reselects the covered ground-truth action.
Continuation recall	Whether the policy recovers the target when it continues the immediately preceding behavior in the trajectory.
Continuation precision	Whether a predicted continuation is correct; this controls for an indiscriminate tendency to repeat the previous action.
Evidence-conditioned accuracy	Whether explicitly represented prior-action evidence separates correct from incorrect selections. This is a descriptive diagnostic, not an intervention.

Table 1: **Operational definition of the reported capability axes.** The metrics isolate selection, predictive-prior retention, error correction, trajectory continuity, and evidence–decision coupling. Their corresponding measurements are consolidated in Figure 1.

the represented evidence discriminates correct from incorrect decisions. Because this split is observational, it motivates—but does not replace—the paired interventions below.

The belief channel provides a complementary control. Answer-only supervision often collapses belief into an action copy, whereas candidate alignment and the final EGO model largely avoid this failure mode. For the candidate-aligned policy, replacing only the belief while holding reasoning fixed yields a paraphrase-adjusted action-flip rate of 9.3%. Low echo and nonzero interventional sensitivity together show that the belief field can carry decision-relevant state without simply restating the answer. This causal interpretation applies to the evaluated candidate-aligned policy only; the final model lacks the corresponding intervention. Table 3 makes the contrast visible in one controlled trace comparison.

What Is and Is Not Established

The current same-cohort evidence establishes three claims. First, candidate-constrained training improves selection beyond both the world-model Top-1 policy and answer-only supervision. Second, the improvement appears in both retention and correction rather than in a single easy subset. Third, the gain is causally dependent on completed-action history. Together these results support the paper's modular account: the world model defines a visually grounded action boundary, while the language policy learns to interpret task and trajectory context within that boundary.

The final training stage augments Retrospection-with-Replay with grounding oversampling at a fixed optimization budget. It restores overall selection to the candidate-aligned level while strengthening retention,

Cooking anticipation case. Observation interval 3620.4–3628.4 s; target time 3631.0 s; 122 completed actions in the available history.

WM candidate set (shuffled; scores and ranks hidden from the policy): check heat (GT, WM rank 3) · flip flatbread · transfer flatbread · cook flatbread · press dough · roll dough · knead dough (WM rank 1) · sprinkle flour · shape dough · cut dough.

Completed-action history (last 8): … → roll dough → flip ingredient → check heat → roll dough → check heat → flip ingredient → press dough → cook flatbread.

Model	reasoning → belief → action
Base	*"The current frame* SHOWS *the flatbread being cooked on the hot surface, and the person's hand is positioned to flip it… Flipping the flatbread is the logical next step to ensure even cooking."* BELIEF — Flipping the flatbread to cook the other side. → **flip flatbread** × [scene description via "shows"]
GT-only	*"The current frame* SHOWS *the flatbread being cooked on the hot surface… The most logical next step, given the context of cooking flatbread, is to flip it… This aligns with <u>the previous actions</u> of checking heat and flipping the flatbread."* BELIEF — **flip flatbread** [action-copying belief] → **flip flatbread** × [scene description via "shows"]
Cand.-CE	*"The current frame* SHOWS *the flatbread being cooked on the hot surface, and <u>the previous action</u> was flipping the ingredient. The next logical step is to continue cooking the flatbread, as it is still on the fire and needs to be cooked through."* BELIEF — Cook the flatbread on the fire. → **cook flatbread** × [scene description via "shows"]
EGO	***"I have just finished pressing the dough and am now observing the fire to assess the heat for cooking. The flames are visible and active, indicating the stove is ready.*** *I need to ensure the heat is suitable before proceeding with the next step in cooking."* BELIEF — I am in the middle of assessing the heat level for cooking the flatbread. → **check heat** ✓ [state-bearing belief; no explicit prior-action reference]

Table 3: **A state-bearing belief enables correction when an action-copying belief does not.** The four evaluated policies receive the same observation, completed-action history, shuffled candidate set, and prompt. Generated traces are reproduced verbatim, with … marking omitted text. To avoid anecdotal selection, we apply a fixed criterion to the common covered set: all four traces must be valid, GT-only must copy its chosen action into the belief field and answer incorrectly, and EGO must answer correctly. This criterion identifies 31 of 915 points; the example shown is one of the 18 for which Cand.-CE is also incorrect. <u>Underlined</u> spans explicitly reference prior actions under the same rule used for the mention-rate analysis, whereas **bold** spans express the model's current task state.

continuation recovery, and evidence utility (Figure 1). However, the preregistered 1 pp non-inferiority criterion is not met: the paired SelAcc difference is -0.64 pp (95% CI $[-3.88, +2.91]$), and correction also declines. The interval contains zero, so the result does not establish degradation; equally, its lower bound is too low to establish preservation under the specified margin. We therefore treat the final stage as a recovery of the aggregate score with a changed capability profile, not as uniformly lossless Retrospection.

Limitation of the final-stage intervention. Axis 5 is not established for the final EGO model because belief swap was not evaluated after grounding oversampling. At the preceding replay stage, belief sensitivity remained below its preregistered threshold while reasoning sensitivity was the strongest among the evaluated conditions. This suggests a redistribution toward the reasoning channel, but the final model's zero belief–action echo is not an interventional substitute. Final-stage belief-utility diagnostics therefore remain exploratory.

Excluded proxy. First-person phrasing is excluded from the capability profile because it changes sharply with prompt regime and does not positively predict correctness within the answer-only control. Contrastive connectives are likewise rare, and their association with accuracy reverses across the replay and grounding stages on a small subset. Both are therefore treated as template-sensitive surface forms rather than evidence of embodied reasoning; neither supports the paper's claims.